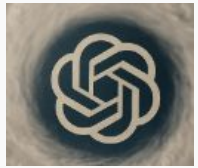
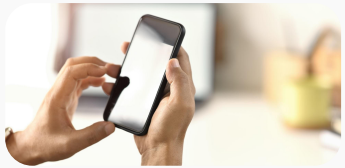
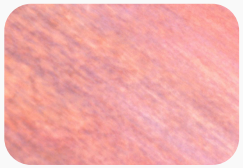
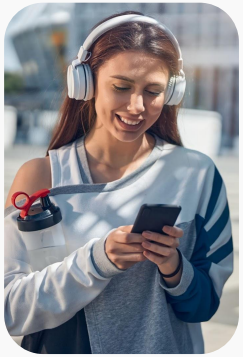
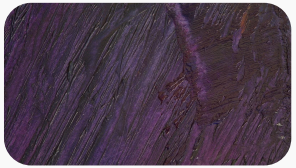
The 5 Kinds of Bots
The NIST/CISA SP800-53 overlay



By Matt Raio MSIS / Author / Founder of Raiolabs LLC

What is the NIST/CISA Overlay?

Recently, the author was invited to participate in a NIST/CISA slack channel to help with the organizations to comment on a yet to be published document that outlines 5 broad categories of artificial intelligence products.

During the summer, I supported an open community that focused on innovating with these bots. There are some very large players like Google that actively worked with a number of our solutions. Once something is put out there in the public, the open community came up with ingenious ways of using them.

My contributions earned me a spot in this slack channel. (to say nothing of name recognition for the 'bot overlord' concept).

After working with the many bots, I've categorized their danger using the meteorological system of categorizing hurricanes. This will help new and inexperienced users to gauge the risks in working with them.

By the end of this session, you will be able to:

- Identify the five key ways AI (often called "bots" or "digital assistants") is being used outside of the education vertical.
- Practice responsible and secure use of these new tools, protecting yourself and the college's information.

1. The Digital Creator

What it is: AI that generates new content, like writing summaries, emails, articles, or even creative works. **Real-world examples:** Tools that draft emails for you, summarize long documents, or help write a powerpoint presentation ;). **Hurricane Category:** Category 1 (Minimal damage, but still impactful!)



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2. The Digital Forecaster

What it is: AI that analyzes data to predict outcomes or make recommendations. **Real-world examples:** Product recommendation engines (e.g., Amazon, Netflix), AI that screens resumes for job applications, or tools that predict market trends. **Hurricane Category:** Category 2 (Moderate damage, requiring more attention.) Think ‘minority report’, the movie…but not as graphic as catching murderers.

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3. The Solo Task Manager

What it is: AI designed to handle one specific, automated job independently. **Real-world examples:** Chatbots for customer service, automated data entry bots, or AI that manages appointment scheduling. **Variation:** Imagine multiple solo task managers working in parallel, each handling a distinct automated job. **Hurricane Category:** Category 3 (Extensive damage, significant protective measures needed.)

Don’t think of this as job replacement. Think of it as automating the tasks that are repetitive to bots that you control. You still are the stake owner and responsible for a number of bots doing your work. The author believes there is an algorithm to calculate the increase in liability when deploying too many bots. Similar to the movie Multiplicity where each new version of “Doug” may cause issues for the original.

As one who came from automation at a large telco, when we fully automated the triaging of 80 percent of our problems from 20 percent of the repeating errors, we had time to design and create new services.

Another analogy is looking at yourself in between two mirrors. They are reflections of you that can go on until you can’t see them anymore. They perform the exact same task as you do 1000x.





4. The Coordinated Team (a.k.a “the bot overlord”)

What it is: Multiple AI systems working together on complex tasks, often with one bot controlling or coordinating others. **Real-world examples:** Advanced robotic systems in manufacturing, smart city management platforms, or sophisticated logistics and supply chain AI. (Note: Using multiple bots to control other bots is currently out of the scope of College's typical usage.) **Hurricane Category:** Category 4 (Catastrophic damage, widespread impact and disruption.)

Originally, I started a podcast on the new technology called **MCP**. That has morphed into /slash commands and /extensions that allow a bot to learn what tool to use and how to perform the task without additional coding.

Then compound the issue with your central bot communicating in real time with other bots and this is a massive problem I told my former cybersecurity students; they will effectively become ‘bot overlords’ in the future. Although I thought it was funny at the time, now there are products on the market that makes it easier to perform complex tasks between the bots.

The Security Builder

What it is: This refers to the role of humans (the "Agentic Engineer") who use bots as co-creators in the process of building and securing AI tools and systems themselves. It's about how we ensure the development of these powerful tools is safe and responsible. **Real-world examples:** A cybersecurity analyst using AI to identify vulnerabilities in code, a developer leveraging AI for secure code generation and testing, or an IT professional deploying AI-powered security monitoring. **Hurricane Category:** Category 5 (Absolute devastation, the highest level of preparedness and understanding is critical!)

This is a little bit of sci-fi where the user forms a working relationship with the bot of their choice. In this case, my choice is Google's Gemini-cli. The bot learned from me as I learned from it. The interior team (Google) learns from my interactions with it.

It does way more than handle just code completion. It literally figures out ways to get around blockers in real time! An example is building a program to interact with IBM's Qiskit Quantum computer where I got to ask two questions of deep concern. Actually, chatgpt recommended the two questions and I passed it along to the quantum program. To which the ***quantum realm responded back to Chatgpt's questions!***

The dilemma for a user is if you are not completely rooted in network fundamentals, this bot will drive the project into an area the user didn't anticipate or worse, you think you're actually talking to a real person. But it's just an echo chamber. So the bot choosing this pic is a bit creepy to me.



Conclusion

Understanding the 5 broad categories that NIST/CISA came up with is going to help the user to start thinking about handling information as you do your day to day tasks.

Leadership can now write a framework to guide in the use of these bots in combination.

The new department can also craft training modules for staff and students based on these categories as defined by NIST/CISA.

Because I'm in the slack channel, I have access to the concept paper and I can share it too.

As an exercise, contemplate on how any of these bots can add value to your services. Can it fix an issue? Can it give time back so you can focus on the students? These are things that can quickly be prototyped in 20 minutes with an Agentic Engineer.

Thank you for attending!

Feel free to contact me for more information at matt.raio@gmail.com

